A sub-20 upper bound for the Kalton–Roberts constant

GPT-5.5 Pro

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Abstract

Let K_{KR} denote the optimal Kalton–Roberts constant for approximately additive real-valued set functions on algebras of sets. Kalton and Roberts proved $K_{\mathrm{KR}} \le 89/2$, and Bondarenko, Prymak and Radchenko improved this to $K_{\mathrm{KR}} \le 38.8$. We prove

$$K_{\mathrm{KR}} \le \frac{4898}{245} = 19.991836734\ldots < 20.$$

The proof keeps the expander-recombination method but uses two pieces of information special to the additive problem. First, the extremal certificate for best approximation by additive measures may be imbalanced; the smaller side has mass $q \le 1/2$. Secondly, triple intersections lower item frequencies from q to q^3 while increasing average deficits by only four. When q is close to $1/2$, a parallel near-minimum construction and a two-sided recombination inequality give the final saving. The required expander families are certified by a Pippenger-style entropy argument with four explicit parameter triples.

1 Introduction and finite reduction

Let Ω be a set and let $\mathcal{F}$ be an algebra of subsets of Ω. A function $f : \mathcal{F} \to \mathbb{R}$ is called Δ-additive if $f(\varnothing) = 0$ and

$$|f(A) + f(B) - f(A \cup B)| \le \Delta \qquad (A, B \in \mathcal{F},\ A \cap B = \varnothing). \tag{1}$$

The Kalton–Roberts theorem asserts that there is a universal constant C such that every Δ-additive f is within $C\Delta$, uniformly on $\mathcal{F}$, of a finitely additive signed measure. We write K_{KR} for the smallest such constant.

Kalton and Roberts proved $K_{\mathrm{KR}} \le 89/2$ in [4]. Bondarenko, Prymak and Radchenko improved the upper bound to 38.8 in [1]. Gnacik, Guzik and Kania proved the lower bound $K_{\mathrm{KR}} \ge 3$ in [3]. Related constants for approximately modular set functions were studied by Feige, Feldman and Talgam-Cohen in [2].

The purpose of this note is to prove the following improved upper bound.

Theorem 1.1. *For every set algebra $\mathcal{F}$ and every Δ-additive function $f : \mathcal{F} \to \mathbb{R}$ there is a finitely additive signed measure μ on $\mathcal{F}$ such that*

$$\sup_{A \in \mathcal{F}} |f(A) - \mu(A)| \le \frac{4898}{245}\,\Delta.$$

Consequently,

$$K_{\mathrm{KR}} \le \frac{4898}{245} < 20.$$

The proof is finite-dimensional. We shall work on 2^U, where U is finite, and then pass to general algebras by compactness. The following standard reduction is included to make the normalization explicit.

Lemma 1.2 (Finite reduction)**.** *Suppose that Theorem 1.1 is known for every finite power set algebra 2^U with a constant C. Then it holds with the same constant C for every set algebra.*

Proof. It is enough to consider $\Delta > 0$, since the case $\Delta = 0$ is exact additivity. By scaling, take $\Delta = 1$. Let $f : \mathcal{F} \to \mathbb{R}$ be 1-additive, and assume the finite theorem with constant C.

For each $A \in \mathcal{F}$ set $I_A = [f(A) - C, f(A) + C]$, and consider the compact product space

$$X = \prod_{A\in\mathcal{F}} I_A.$$

For disjoint $A, B \in \mathcal{F}$, the equation $\nu(A \cup B) = \nu(A) + \nu(B)$ defines a closed subset of X. We claim that the family of all these closed additivity conditions has the finite intersection property. Indeed, take finitely many such equations and finitely many coordinates. They involve only finitely many sets of $\mathcal{F}$, hence are contained in a finite subalgebra $\mathcal{F}_0$. Identifying $\mathcal{F}_0$ with a power set of its atoms, the finite theorem gives an additive signed measure ν_0 on $\mathcal{F}_0$ with $|\nu_0(A) - f(A)| \le C$ for all $A \in \mathcal{F}_0$. Extend ν_0 arbitrarily to a point of X on the other coordinates. This point satisfies the chosen finite additivity equations and all relevant coordinate bounds.

Compactness therefore gives a point $\mu \in X$ satisfying every finite-additivity equation. Thus μ is a finitely additive signed measure on $\mathcal{F}$, and $\mu \in X$ gives $|f(A) - \mu(A)| \le C$ for every $A \in \mathcal{F}$. □

Henceforth U is finite and $\Delta = 1$. A set function $\mu : 2^U \to \mathbb{R}$ is additive if

$$\mu(S) = \sum_{i\in S} a_i$$

for some real coefficients $(a_i)_{i\in U}$. Let L denote the subspace of additive functions.

2 Dual certificates and low-frequency collections

Let $f : 2^U \to \mathbb{R}$ be 1-additive. Subtract a closest additive function from f. This preserves (1), preserves $f(\varnothing) = 0$, and allows us to assume that the zero function is a closest additive approximation. Put

$$M := \|f\|_\infty = \operatorname{dist}(f, L).$$

If $M = 0$ there is nothing to prove, so assume $M > 0$.

Lemma 2.1 (Dual certificate)**.** *There is a signed measure λ on 2^U such that*

(i) $\|\lambda\|_1 = 1$*;*

(ii) $\sum_{S\ni i} \lambda_S = 0$ *for every $i \in U$;*

(iii) $\lambda_S > 0$ *only when $f(S) = M$, and $\lambda_S < 0$ only when $f(S) = -M$.*

Moreover, λ may be chosen with rational masses.

Proof. Since 0 is a best approximation to f from the finite-dimensional subspace $L \subset \ell_\infty(2^U)$, the Hahn–Banach separation criterion for best approximation gives a functional $\lambda \in L^\perp \subset \ell_1(2^U)$ with $\|\lambda\|_1 = 1$ and

$$\langle \lambda, f \rangle = \|f\|_\infty = M.$$

The equations defining $L^\perp$ are exactly

$$\sum_{S \ni i} \lambda_S = 0 \qquad (i \in U).$$

Equality in Hölder's inequality forces λ to be supported on points at which $|f(S)| = M$, with the sign of λ_S agreeing with the sign of $f(S)$. In particular $\lambda_\varnothing = 0$, because $f(\varnothing) = 0$ and $M > 0$.

For rationality, keep only the active sets on which such a certificate may be nonzero. The conditions of vanishing item marginals and prescribed signs form a homogeneous rational polyhedral cone. Since this cone contains a nonzero real point, it contains a nonzero rational point. Normalizing its ℓ_1 norm gives the rational certificate. $\square$

Write

$$p := \lambda^+(2^U), \qquad q := \lambda^-(2^U), \qquad p + q = 1.$$

Replacing f by $-f$, if necessary, assume

$$q \le p, \qquad q \le \frac{1}{2}. \tag{2}$$

Both signs occur in the certificate: if, say, $q = 0$, then the non-negative vector λ has all item marginals equal to zero and no mass at $\varnothing$, which is impossible unless $\lambda = 0$. Thus $p, q > 0$.

Let $u := f(U)$. For a set A with $f(A)$ near M, write its deficit as $M - f(A)$. For a set B with $f(B)$ near $-M$, write its surplus as $M + f(B)$.

Lemma 2.2. *We have $-1 \le u \le 1$. There are two rationally weighted collections of subsets of U with the following properties.*

(a) *A near-maximum collection $\mathcal{A}$ of total weight 1, item frequency q, and average deficit*

$$d_{\mathcal{A}} := \mathbb{E}_{A \in \mathcal{A}}(M - f(A)) \le q(1 - u).$$

(b) *A near-minimum collection $\mathcal{B}$ of total weight 1, item frequency p, and average surplus*

$$s_{\mathcal{B}} := \mathbb{E}_{B \in \mathcal{B}}(M + f(B)) \le p(1 + u).$$

Here "item frequency α" means that every $i \in U$ belongs to an α-fraction of the weighted collection.

Proof. If P is in the positive support of λ, then $f(P) = M$. Since P and P^c are disjoint and $P \cup P^c = U$,

$$M + f(P^c) - u \in [-1, 1].$$

As $f(P^c) \ge -M$, this implies $u \ge -1$. Similarly, if N is in the negative support, then $f(N) = -M$ and

$$-M + f(N^c) - u \in [-1, 1].$$

As $f(N^c) \le M$, this implies $u \le 1$.

Let m_i^+ and m_i^- denote the positive and negative λ-marginal masses of item i. Lemma 2.1 gives $m_i^+ = m_i^-$ for every i.

Define $\mathcal{A}$ by taking all positive active sets P with weight λ_P^+ and all complements N^c of negative active sets with weight λ_N^-. Its total weight is $p+q=1$. The frequency of item i in $\mathcal{A}$ is

$$m_i^+ + (q - m_i^-) = q.$$

The positive active sets have deficit 0. For a negative active set N,

$$-M + f(N^c) - u \in [-1, 1],$$

so $f(N^c) \geq M + u - 1$, and hence $M - f(N^c) \leq 1 - u$. This proves the estimate for $d_{\mathcal{A}}$.

Define $\mathcal{B}$ by taking all negative active sets N with weight λ_N^- and all complements P^c of positive active sets with weight λ_P^+. Its item frequency is

$$m_i^- + (p - m_i^+) = p.$$

The negative active sets have surplus 0. For a positive active set P,

$$M + f(P^c) - u \in [-1, 1],$$

so $f(P^c) \leq -M + u + 1$, and hence $M + f(P^c) \leq 1 + u$. This proves the estimate for $s_{\mathcal{B}}$. □

Since the weights may be chosen rational, each weighted collection used below may be replaced by an ordinary finite multiset after duplicating sets. We shall use this without further comment.

3 Intersections and recombination

The next elementary observation is where additivity gives more than the general weakly modular formulation.

Lemma 3.1 (Approximate modularity). *For arbitrary $A, B \subseteq U$,*

$$|f(A) + f(B) - f(A \cup B) - f(A \cap B)| \leq 2.$$

Consequently, if A and B have deficits a and b, then $A \cap B$ has deficit at most $a + b + 2$. If A and B have surpluses a and b, then $A \cap B$ has surplus at most $a + b + 2$.

Proof. Let $I = A \cap B$ and $Y = B \setminus A$. Then $B = I \sqcup Y$ and $A \cup B = A \sqcup Y$, with both unions disjoint. Hence

$$f(B) = f(I) + f(Y) + \epsilon_1, \qquad f(A \cup B) = f(A) + f(Y) + \epsilon_2,$$

where $|\epsilon_1|, |\epsilon_2| \leq 1$. Subtracting gives

$$f(A) + f(B) - f(A \cup B) - f(A \cap B) = \epsilon_1 - \epsilon_2,$$

and the first assertion follows.

If $f(A) = M - a$ and $f(B) = M - b$, then

$$f(A \cap B) \geq f(A) + f(B) - f(A \cup B) - 2 \geq M - a - b - 2,$$

using $f(A \cup B) \leq M$. This is the deficit statement. The surplus statement follows similarly: if $f(A) = -M + a$ and $f(B) = -M + b$, then

$$f(A \cap B) \leq f(A) + f(B) - f(A \cup B) + 2 \leq -M + a + b + 2,$$

using $f(A \cup B) \geq -M$. □

For a weighted collection $\mathcal{C}$, let $\mathcal{C}^{\cap 3}$ denote the product collection of all triple intersections $C_1 \cap C_2 \cap C_3$, where the three sets are chosen independently from $\mathcal{C}$ according to its weights.

Corollary 3.2. *The collection $\mathcal{A}^{\cap 3}$ has item frequency q^3 and average deficit at most*

$$D_{\mathcal{A}} \leq 3q(1-u) + 4.$$

The collection $\mathcal{B}^{\cap 3}$ has item frequency p^3 and average surplus at most

$$S_{\mathcal{B}} \leq 3p(1+u) + 4.$$

Proof. Item frequencies multiply under independent intersections. Iterating Lemma 3.1, the deficit of a triple intersection is at most the sum of the three deficits plus 4, and the same holds for surplus. Taking expectations and using Lemma 2.2 proves the two estimates. □

We next recall the recombination device. For $\alpha, r, \theta > 0$ with $\alpha < \theta < 1$, an (α, r, θ)-expander is a bipartite multigraph $G = (V, W; E)$ with

$$|V| = 2k, \qquad |W| = 2\theta k,$$

left degree r and right degree r/θ, such that every subset of V of size at most $2\alpha k$ has at least the same number of distinct neighbours in W. We say that such expanders exist if they exist for all sufficiently large admissible k, meaning that the displayed cardinalities and degrees are integral.

Lemma 3.3 (One-sided recombination)**.** *Suppose that (α, r, θ)-expanders exist. Let $\mathcal{C}$ be a finite multiset of subsets of U with item frequencies at most α and average deficit at most D. Then recombination by such an expander gives a target multiset with item frequencies at most α/θ. If D' is its average deficit, then*

$$(1-\theta)M \leq D - \theta D' + 2r - 1 - \theta. \tag{3}$$

Consequently,

$$M \leq \frac{D + 2r - 1 - \theta}{1-\theta}. \tag{4}$$

Proof. Duplicate the source multiset if necessary so that its size is $2k$ and an (α, r, θ)-expander $G = (V, W; E)$ is available. Align the source sets C_v with the vertices $v \in V$. For each item $i \in U$, the set of source vertices whose sets contain i has size at most $2\alpha k$. Hall's theorem, applied to this vertex set and the expander-neighborhood condition, assigns these occurrences injectively to neighboring vertices in W. Choose one witnessing edge for each assigned pair. Thus each source set C_v is partitioned among the edges leaving v, and the pieces entering a fixed $w \in W$ are disjointly recombined into a target set T_w. Since each item is sent to at most $2\alpha k$ target vertices and the target multiset has size $2\theta k$, the target item frequencies are at most α/θ.

Let C_e be the piece placed on edge e. Since C_v is the disjoint union of its outgoing pieces,

$$\sum_{e \in E(v,W)} f(C_e) \geq f(C_v) - (\deg v - 1).$$

Since T_w is the disjoint union of its incoming pieces,

$$\sum_{e \in E(V,w)} f(C_e) \leq f(T_w) + (\deg w - 1).$$

Summing over all vertices gives

$$2k(M - D) - 2k(r-1) \leq 2\theta k(M - D') + 2k(r - \theta),$$

because $\sum_{v \in V}(\deg v - 1) = 2k(r-1)$ and $\sum_{w \in W}(\deg w - 1) = 2k(r - \theta)$. Dividing by $2k$ and rearranging proves (3). Since $D' \geq 0$, (4) follows. □

Averaging the deficit inequality with its analogue for $-f$ gives the two-sided form needed in the second case.

Lemma 3.4 (Two-sided recombination)**.** *Suppose that (α, r, θ)-expanders exist. Let $\mathcal{C}_+$ be a multiset with item frequencies at most α and average deficit at most D, and let $\mathcal{C}_-$ be a multiset with item frequencies at most α and average surplus at most S. Recombination gives target multisets with item frequencies at most α/θ. Let D' and S' be their average deficit and average surplus. Then*

$$(1-\theta)M \le \frac{D+S-\theta(D'+S')}{2} + 2r - 1 - \theta. \tag{5}$$

Proof. Apply Lemma 3.3 to $\mathcal{C}_+$. Apply the same argument to $-f$ and $\mathcal{C}_-$, observing that surplus for f is deficit for $-f$. Average the two inequalities. The target-frequency assertion follows from the one-sided recombination construction. □

4 The four expander families

We need four explicit expander families. Their existence follows from the standard Pippenger random biregular construction. We include the verification because the final constant depends on the numerical parameters.

For $a > b > 0$ set

$$h(a, b) = a \log a - b \log b - (a-b)\log(a-b).$$

For an integer $r \ge 3$ and $\theta = r/c$, where $c > r$ is an integer, define

$$\Phi_{r,\theta}(x) = h(1, x) + h(\theta, x) + h(rx/\theta, rx) - h(r, rx). \tag{6}$$

Lemma 4.1 (Pippenger entropy check)**.** *Let $0 < \alpha < \theta < 1$, let $r \ge 3$ be an integer, and let $\theta = r/c$ for an integer $c > r$. Suppose that there is $\delta \in (0, \alpha)$ such that*

(i) $e^2\theta^{1-r}\delta^{r-2} < 1$*;*

(ii) $\Phi_{r,\theta}(x) < 0$ *for every* $x \in [\delta, \alpha]$.

Then (α, r, θ)-expanders exist.

Proof. Let $N = 2k$, with N sufficiently divisible. Form a random bipartite multigraph by taking r copies of each of the N left vertices and $c = r/\theta$ copies of each of the θN right vertices, and then choosing a uniformly random perfect matching between the two sets of copies.

Fix $m = xN$ left vertices. If their neighborhood has size less than m, then it is contained in some specified set of m right vertices. The probability of containment in one specified set is at most

$$\frac{\binom{cm}{rm}}{\binom{rN}{rm}}.$$

Thus the expected number of bad left sets of size m is at most

$$\binom{N}{m}\binom{\theta N}{m}\frac{\binom{cm}{rm}}{\binom{rN}{rm}}. \tag{7}$$

For $x \in [\delta, \alpha]$, Stirling's formula, uniformly on this compact interval, gives

$$(7) = \exp\bigl(N\Phi_{r,\theta}(x) + o(N)\bigr).$$

By assumption this is exponentially small, uniformly over $x \in [\delta, \alpha]$, and summing over m in this range still gives a quantity tending to zero.

For $x < \delta$, the cruder estimate

$$\binom{N}{m}\binom{\theta N}{m}\Big(\frac{m}{\theta N}\Big)^{rm} \le \big(e^2\theta^{1-r}x^{r-2}\big)^m$$

applies. Put $\rho = e^2\theta^{1-r}\delta^{r-2} < 1$. For each fixed m, the sharper version of the same estimate is $O(N^{-(r-2)m})$, hence tends to zero. Uniformly for $m < \delta N$, the displayed bound is at most ρ^m. Taking a fixed initial segment and then a geometric tail shows that the small-size contribution also tends to zero. Hence, with positive probability, no bad set of size at most αN exists. □

For the four parameter triples below, take $\delta = 1/100$. The following identities are useful for checking condition (ii):

$$\Phi_{r,\theta}(x) = h(\theta, x) - (r-1)h(1, x) + L_{r,\theta}x,$$

where

$$L_{r,\theta} = \frac{r}{\theta}\log\frac{r}{\theta} - r\log r - \big(\frac{r}{\theta} - r\big)\log\big(\frac{r}{\theta} - r\big),$$

and

$$\Phi''_{r,\theta}(x) = \frac{r-2}{x} + \frac{r-1}{1-x} - \frac{1}{\theta - x}. \tag{8}$$

On $[\delta, \alpha]$ we use the simple estimate

$$\Phi''_{r,\theta}(x) \ge \frac{r-2}{\alpha} - \frac{1}{\theta-\alpha},$$

since the middle term in (8) is positive. For every row in Table 1, the two endpoint values are $< -1/200$ and this lower bound for Φ'' is > 3. Therefore Φ is convex and negative on the whole interval. The final column uses the elementary bound $e^2 < 8$, so condition (i) is also verified. The companion script `verify_kalton.py` checks these inequalities by exact rational interval arithmetic and recomputes the final rational constants.

Name	α	r	θ	endpoint max	Φ'' lower bd.	small-x bd.
E_1	1000/9261	4	4/11	< -0.02706	> 14.61	< 0.0167
E_2	2750/9261	4	4/7	< -0.00961	> 3.09	< 0.0043
E_3	1331/9261	4	2/5	< -0.00651	> 10.01	< 0.0126
E_4	6655/18522	5	5/9	< -0.02982	> 3.25	< 0.000084

Table 1: The four expander certificates. Here $\delta = 1/100$.

5 Proof of the main theorem

We prove Theorem 1.1 for finite U and $\Delta = 1$. By Lemma 1.2 and scaling, this implies the theorem in full generality.

Let $q \le 1/2$ be as in (2), and set

$$q_0 = \frac{10}{21}.$$

We split into two cases.

Case 1: $q \le q_0$

By Corollary 3.2, the multiset $\mathcal{A}^{\cap 3}$ has item frequency at most

$$q_0^3 = \frac{1000}{9261}$$

and average deficit at most

$$D \le 3q(1-u) + 4 \le 6q + 4 \le 6q_0 + 4 = \frac{48}{7},$$

where we used $u \ge -1$.

Apply the expander E_1, namely

$$\left(\frac{1000}{9261}, 4, \frac{4}{11}\right).$$

Let D' be the target average deficit. Lemma 3.3 gives

$$M \le \frac{1039}{49} - \frac{4}{7}D'. \tag{9}$$

The target item frequencies are at most

$$\frac{1000/9261}{4/11} = \frac{2750}{9261},$$

so the expander E_2, namely

$$\left(\frac{2750}{9261}, 4, \frac{4}{7}\right),$$

may be applied to the target multiset. Ignoring the second-generation target deficit in Lemma 3.3 yields

$$M \le 15 + \frac{7}{3}D'. \tag{10}$$

Combining (9) and (10),

$$M \le \sup_{D' \ge 0} \min\left\{\frac{1039}{49} - \frac{4}{7}D', 15 + \frac{7}{3}D'\right\} = \frac{1219}{61} < 20.$$

Case 2: $q \ge q_0$

Now $p = 1 - q \le 1 - q_0 = 11/21$. The deficit multiset $\mathcal{A}^{\cap 3}$ has item frequency $q^3 \le 1/8$, while the surplus multiset $\mathcal{B}^{\cap 3}$ has item frequency

$$p^3 \le \left(\frac{11}{21}\right)^3 = \frac{1331}{9261}.$$

Since $1/8 < 1331/9261$, both item frequencies are at most $1331/9261$.

By Corollary 3.2, the average deficit D of $\mathcal{A}^{\cap 3}$ and the average surplus S of $\mathcal{B}^{\cap 3}$ satisfy

$$\begin{aligned}\frac{D+S}{2} &\le \frac{3q(1-u) + 4 + 3p(1+u) + 4}{2} \\ &= \frac{3(1 + (p-q)u) + 8}{2} \le 7 - 3q \le 7 - 3q_0 = \frac{39}{7}.\end{aligned}$$

Here we used $p \ge q$ and $u \le 1$.

Apply the expander E_3, namely

$$\left(\frac{1331}{9261}, 4, \frac{2}{5}\right).$$

Let

$$X = \frac{D' + S'}{2}$$

be half the sum of the target average deficit and target average surplus. Lemma 3.4 gives

$$M \leq \frac{142}{7} - \frac{2}{3}X. \tag{11}$$

The target item frequencies are at most

$$\frac{1331/9261}{2/5} = \frac{6655}{18522},$$

so the expander E_4, namely

$$\left(\frac{6655}{18522}, 5, \frac{5}{9}\right),$$

may be applied to both target multisets. Ignoring the second-generation target terms in Lemma 3.4 gives

$$M \leq 19 + \frac{9}{4}X. \tag{12}$$

Combining (11) and (12),

$$M \leq \sup_{X \geq 0} \min \left\{ \frac{142}{7} - \frac{2}{3}X, 19 + \frac{9}{4}X \right\} = \frac{4898}{245} < 20.$$

The two cases show $M \leq 4898/245$ after subtracting the closest additive function. Undoing this normalization proves the finite theorem. Lemma 1.2 and scaling complete the proof of Theorem 1.1.

6 Remarks on verification

The proof above is intentionally conservative about expander parameters. The parameter $q_0 = 10/21$ is chosen so that the first case uses the one-sided triple-intersection argument up to frequency $(10/21)^3$, while the second case keeps both the near-maximum and near-minimum triple-intersection collections below the third expander threshold. The final constants are exact rational balances of two affine inequalities in each case.

The only numerical part is Table 1. It reduces to checking finitely many logarithmic inequalities at rational points, together with the rational lower bound (8). The companion verification file implements the identity

$$\log z = 2 \sum_{n \geq 0} \frac{t^{2n+1}}{2n+1}, \qquad t = \frac{z-1}{z+1},$$

with range reduction to a fixed interval around 1 and a geometric upper bound on the positive tail. Thus all interval comparisons in the script are made with exact rational arithmetic.

References

[1] A. V. Bondarenko, A. Prymak, and D. Radchenko. On concentrators and related approximation constants. *J. Math. Anal. Appl.*, 402(1):234–241, 2013. doi:10.1016/j.jmaa.2013.01.019.

[2] U. Feige, M. Feldman, and I. Talgam-Cohen. Approximate modularity revisited. *SIAM J. Comput.*, 49(1):67–97, 2020. doi:10.1137/18M1173873.

[3] M. Gnacik, M. Guzik, and T. Kania. Approximate modularity: Kalton's constant is not smaller than 3. *Proc. Amer. Math. Soc.*, 149(2):661–669, 2021. doi:10.1090/proc/15195.

[4] N. J. Kalton and J. W. Roberts. Uniformly exhaustive submeasures and nearly additive set functions. *Trans. Amer. Math. Soc.*, 278(2):803–816, 1983. doi:10.1090/S0002-9947-1983-0701524-4.

[5] N. Pippenger. Superconcentrators. *SIAM J. Comput.*, 6(2):298–304, 1977. doi:10.1137/0206022.